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UnravelNet: A backbone for enhanced multi-scale and low-quality feature extraction in remote sensing object detection

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ABSTRACT

Remote sensing object detection (RSOD) faces significant challenges, including substantial variations in object sizes within a single image and the presence of low-quality object features. Previous approaches have attempted to mitigate these issues by employing large spatial receptive fields, such as through large kernel convolutions, or by using multi-scale convolutional kernels to capture features from objects of varying sizes. However, large kernel convolutions often fail to extract detailed object features effectively, while multi-scale convolutional kernels struggle to capture relevant features of low-quality or blurry objects, leading to reduced detection accuracy. To address these issues, this study presents UnravelNet, a novel backbone network tailored for RSOD. UnravelNet innovatively disentangles the coupling of multi-scale features through a non-expansive multi-branch attention mechanism, incorporating point, local and mid-range attention. Additionally, UnravelNet improves the representation of low-quality features by integrating edge information and Gaussian distributions. This approach enables UnravelNet to efficiently utilize a limited number of parameters while effectively extracting both multi-scale and low-quality features. Extensive experiments on standard RSOD benchmarks demonstrate the superior performance of UnravelNet, achieving state-of-the-art results with mAPs of 79.99% on DOTA-v1.0, 73.09% on DOTA-v1.5, 69.48% on DIOR-R and 44.44% on FAIR1M-v1.0. These results highlight the improvements in both accuracy and efficiency for RSOD tasks, underscoring the potential of UnravelNet as a robust solution for remote sensing applications. The code will be available at https://github.com/lwCVer/UnravelNet.

1. Introduction

Remote sensing object detection (RSOD) is a critical subfield of computer vision (CV) that has gained considerable attention due to its unique challenges. Unlike traditional object detection tasks that focus primarily on terrestrial images, RSOD processes images captured from satellites or aerial platforms. These images show considerable variation in object sizes and frequently include features of low-quality objects. Such characteristics complicate the direct application of conventional CV techniques, necessitating the development of specialized algorithms and architectures specifically for remote sensing (RS) tasks.

To better match the features of various objects and improve detection accuracy, anchor-based methods (Ding et al., 2019; Wang et al., 2021a; Cheng et al., 2022b) utilize predefined anchors to match objects, demonstrating high positioning accuracy and stability. These anchor boxes provide prior knowledge for detection tasks, helping the model more accurately locate the object bounding boxes. Anchor-free methods (Li et al., 2022; Cheng et al., 2022a) directly predict the object center or key points, avoiding the complex matching process between anchors and objects, showing significant flexibility. Additionally, some methods focus on improving the representation of oriented bounding boxes (Xu et al., 2020; Xie et al., 2021; Yao et al., 2022), designing new loss functions (Yang and Yan, 2020; Yang et al., 2021b,c, 2023; Qian et al., 2023), or adaptive alignment of convolutional features (Han et al., 2021a; Yang et al., 2021a) to further enhance detection performance. Despite these advancements, feature extraction

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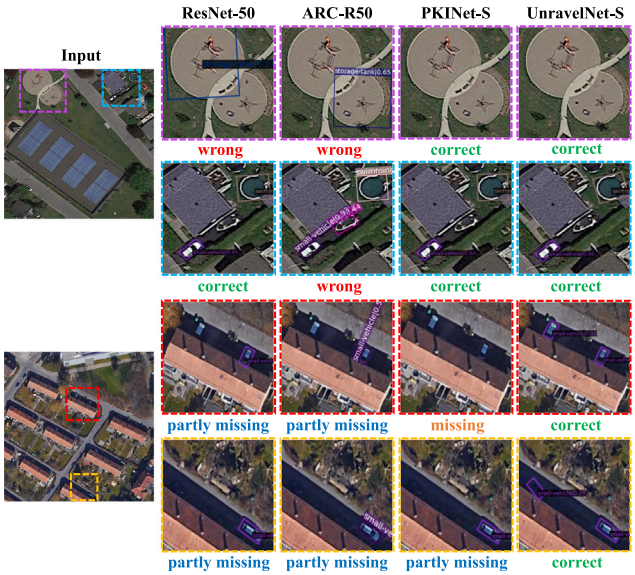

Fig. 1. Visualization results of different backbones under the Oriented RCNN (Xie et al., 2021) framework on DOTA (Xia et al., 2018). Our UnravelNet demonstrates strong detection capabilities under multi-scale and low-quality conditions.

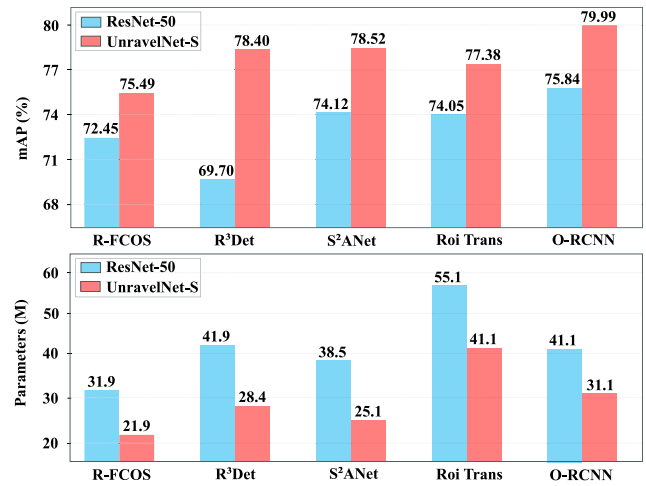

Fig. 2. Performance comparison of UnravelNet backbone with various decoders on the DOTA-v1.0 test set (Xia et al., 2018). Top: mAP comparison. Bottom: Parameter comparison.

methods specifically designed for RSOD have yet to be effectively explored.

To address these challenges, some recent methods have attempted to dynamically adjust large receptive fields, such as LSKNet (Li et al., 2023b) using large kernel convolutions. LSKNet seeks to capture contextual information over a large area to detect objects of varying sizes. However, this approach has limitations: large kernel convolutions struggle to extract detailed object features due to their coarse resolution. Furthermore, the application of large kernels to small objects in RS images is inefficient, leading to missed detections.

Another emerging approach involves multi-scale convolutional kernels, exemplified by PKINet (Cai et al., 2024). PKINet uses kernels of varying sizes to extract features from objects at different scales, with the goal of improving detection across resolutions. While promising, this method still faces challenges in effectively extracting features from low-quality or blurry objects, which diminishes detection precision. This highlights the need for more efficient techniques that can robustly handle low-quality object features. As shown in Fig. 1, some previous methods such as ResNet-50 (He et al., 2016) and ARC-R50 (Pu et al., 2023) have limitations in handling multi-scale object features (upper part of Fig. 1) and low-quality object features (lower part of Fig. 1), which are prone to false detection or missed detection, resulting in suboptimum detection performance. Although the current state-of-the-art (SOTA) method, PKINet, extracts object features at different scales through multi-scale convolutional kernels, providing a new approach to address the challenge of multi-scale objects, it ineffectively extracts features of low-quality or blurred objects.

To overcome these limitations, this article presents UnravelNet, a novel backbone network specifically designed for RSOD. UnravelNet introduces an innovative architecture that unravels feature coupling through a non-expansive multi-branch attention mechanism. This approach enables more accurate and efficient feature extraction, particularly for multi-scale and low-quality objects.

As illustrated in Fig. 1, compared with current methods, UnravelNet effectively handles both multi-scale and low-quality object features, greatly improving detection capability. Meanwhile, as shown in Fig. 2, UnravelNet could be seamlessly integrated into various detection decoders. Despite having a more compact structure compared to ResNet-50, UnravelNet significantly enhances accuracy, demonstrating its potential for improving RSOD performance.

UnravelNet incorporates four distinct attention mechanisms: point attention (PA), local attention (LA), sparse mid-range attention (SMRA), and low-quality feature enhancement attention (LFEA). Each mechanism is tailored to enhance different aspects of feature extraction:

PA focuses on individual points in the feature map, improving the localization of object features by integrating channel information, which enhances the detectability of small and detailed objects.

LA captures the surrounding region of each point, capturing local contextual information that helps in detecting general regular objects. By aggregating features from neighboring points, LA provides a more comprehensive representation of object sizes and shapes, which allows the model to better understand and process details and structures in visual data.

SMRA bridges the gap between local and global attention, extending the receptive field without increasing parameter complexity and FLOPs. It allows UnravelNet to effectively capture mid-range context and improve feature expression for irregular objects.

LFEA significantly improves the representation of low-quality and blurry objects by reducing noise, representing object boundaries, and modeling uncertainty in feature extraction, which improves the network's ability to handle ambiguous cases.

It is pertinent to note that RS images consist primarily of a large number of objects of varying sizes, but there are few giant objects in a single image. Thus, the establishment of global attention in RS images remains to be discussed and poses higher challenges for edge devices and mobile terminals due to the increased complexity. Therefore, this aspect of RSOD is not explored further in this work.

UnravelNet enhances the feature representation of multi-scale objects by PA, LA and SMRA, and enhances the representation of low-quality features by LFEA. These enhancements are crucial for robust detection in challenging visual conditions. The multi-branch attention mechanism in UnravelNet allows for efficient use of parameters while extracting multi-scale features. By separating the feature extraction process into multiple branches, UnravelNet can not only focus on different aspects of the image but also leverage the inherent redundancy of features. This enables the model to capture richer and more discriminative features, leading to more accurate and robust RSOD.

Extensive experiments on standard RSOD benchmarks demonstrate the superior performance of UnravelNet. It achieved SOTA results on four RSOD datasets: DOTA-v1.0 (Xia et al., 2018) (79.99% mAP), DOTA-v1.5 (Xia et al., 2018) (73.09% mAP), DIOR-R (Cheng et al.,

2022a) (69.48% mAP) and FAIR1M-v1.0 (Sun et al., 2022) (44.44% mAP). These results highlight the effectiveness of the UnravelNet architecture in addressing the challenges in RSOD. The combination of multi-branch attention mechanisms and feature enhancement techniques positions UnravelNet as a promising solution for RSOD, paving the way for further advancements in RS applications.

The primary contributions are the following:

- A novel multi-branch attention is introduced, incorporating four distinct attention mechanisms that enhance the detection and precision of features across varying conditions (multi-scale and low-quality) in RS imagery.
- The UnravelNet backbone network is presented, specifically designed for RSOD, significantly improving the representation of RS images and enabling better object delineation and uncertainty management.
- Empirical evaluation on multiple RSOD benchmark datasets demonstrates the superior performance of the UnravelNet backbone, achieving top scores on several datasets and confirming its potential as a leading solution in RSOD tasks.

2. Related work

Recent advancements in object detection have significantly enhanced performance across various domains. However, RSOD presents unique challenges that distinguish it from object detection in natural images. These challenges include the wide variety of object scales, dense object arrangements, arbitrary orientations, and lower-quality features typical of RS images.

Traditional methods that employ horizontal bounding boxes (HBB) for detecting rotated objects in RS images often introduce excessive noise and may group multiple objects within a single anchor, complicating model training and convergence. Consequently, the adoption of oriented bounding boxes (OBB) for RSOD has gained prominence.

Several RSOD methods have been proposed to address these challenges. For example, methods such as those by Yang et al. (2018) and Azimi et al. (2018) design anchors with varying scales, aspect ratios and rotation angles at each position on the feature map. Although comprehensive, these approaches are computationally intensive and generate a large number of negative samples. RoI Transformer (Ding et al., 2019) addresses this issue by learning and predicting rotated regions of interest (RoIs) from horizontal anchors, thereby reducing the need for predefined anchors. Further developments, such as S^2ANet (Han et al., 2021a) and R^3Det (Yang et al., 2021a), employ feature alignment techniques to improve prediction precision. However, these methods still encounter challenges, such as discontinuities at object boundaries. To overcome these issues, Gliding Vertex (Xu et al., 2020) employs a sliding vertex approach that learns offsets for the four corners of a HBB, allowing for the localization of arbitrarily oriented objects. However, this method continues to rely on horizontal RoIs for classification and regression, which may cause misalignment between objects and features. Oriented R-CNN (Xie et al., 2021) adopts a midpoint offset strategy to predict the offsets of midpoints of each side of the bounding box in the oriented region proposal network (RPN), generating a six-parameter OBB for detection. Oriented RepPoints (Li et al., 2022) explore an alternative approach by detecting objects using adaptive points in RS images.

In addition, several studies have focused on developing loss functions tailored for OBB regression. CSL (Yang and Yan, 2020) introduces circular smooth labels, transforming the angle regression task into an angle classification task to address the convergence issues caused by the angular periodicity inherent in traditional Ln-norm losses. SCRDet (Yang et al., 2019) employs IoU-smooth L1 loss, applying IoU weighting to suppress out-of-bound angles. GWD (Yang et al., 2021b) and KLD (Yang et al., 2021c) approximate SkewIoU using Gaussian distribution distance measures. KFIoU (Yang et al., 2023) offers a simpler and more efficient approximation based on Gaussian distribution and Kalman filtering, while FRIoU (Qian et al., 2023) leverages a unified transferring strategy to adapt horizontal bounding box regression losses for oriented detection and incorporates a focal mechanism to prioritize challenging samples.

Designing a dedicated backbone network for different tasks is also very important. ViT (Dosovitskiy et al., 2020) demonstrate that a pure transformer architecture, applied directly to sequences of image patches, can achieve SOTA performance on general image classification tasks when pre-trained on larger datasets, which has attracted widespread attention. SST (Chen et al., 2021) effectively captures the global context of objects with diverse orientations and scales by introducing a dual-path transformer architecture and a sparse token sampler, significantly enhancing the accuracy and efficiency of building extraction. LWGANet (Lu et al., 2025) proposes a lightweight backbone network that achieves multi-scale feature modeling while reducing computational complexity. It attains SOTA performance on 12 datasets spanning four RS visual tasks, demonstrating a significant balance between efficiency and accuracy in RS image analysis. RSMamba (Chen et al., 2024) and RSM (Zhao et al., 2024), both based on state space model (SSM) (Gu et al., 2021), efficiently handle RS image classification (Chen et al., 2023b) and dense prediction tasks (Liu et al., 2024), respectively. MambaRetinaNet (Chen et al., 2025a) integrates CNN and Mamba for efficient multi-scale RSOD. DMM (Zhou et al., 2025) employs a disparity-guided multispectral Mamba framework for oriented object detection. LS-MambaNet (Yan et al., 2025) combines large strip convolution with Mamba for enhanced detection of high-aspect-ratio targets in RS images. DynamicVis (Chen et al., 2025b) presents a dynamic visual perception foundation model for RS visual tasks, utilizing dynamic visual perception mechanisms and SSM to solve the problem of high computational complexity in processing high-resolution images.

To better handle the challenges such as large object scale variations in RSOD, ARC (Pu et al., 2023) adaptively rotates convolutional kernels to extract features from objects with varying orientations, using a conditional computation mechanism to accommodate arbitrary object orientations. LSKNet (Li et al., 2023b) dynamically adjusts receptive field sizes to provide the necessary contextual information for objects at varying scales. PKINet (Cai et al., 2024) leverages multi-scale convolution kernels to effectively capture local contextual information and multi-scale features.

In this article, we propose a novel feature extraction backbone UnravelNet to address the vast variation in object scales and the prevalence of low-quality features in RS images. The innovation of UnravelNet lies in two main aspects: first, it replaces large-kernel and multi-scale convolutions with a non-expansive multi-branch attention scheme, enabling the extraction of multi-scale texture features across varying receptive fields; second, it enhances the representation of low-quality features by incorporating edge information and Gaussian distribution. These innovations work together to extract both low-quality features and multi-scale information, leading to improved performance in RSOD tasks.

3. Methodology

This section introduces the architecture and essential components of the proposed UnravelNet backbone, explicitly designed to address the challenges of multi-scale and low-quality feature extraction in RSOD tasks.

3.1. Architectural overview

As illustrated in Fig. 3, The architecture of UnravelNet is organized into four progressive stages, with downsampling ratios of 1/4, 1/8, 1/16, and 1/32, respectively. The network is composed of a stem layer,

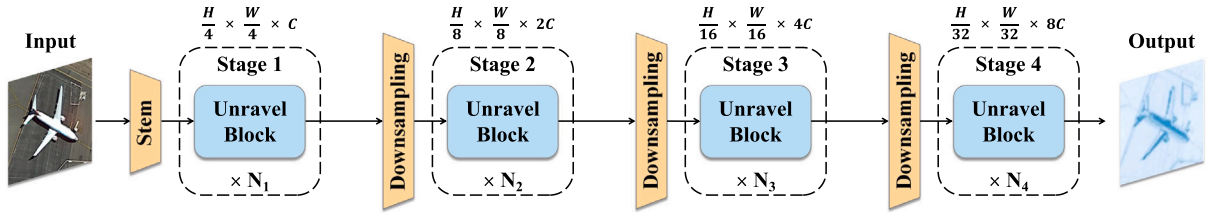

Fig. 3. Overview of UnravelNet architecture. It encompasses four stages, each featuring dimensions of $\frac{H}{4} \times \frac{W}{4} \times C$, $\frac{H}{8} \times \frac{W}{8} \times 2C$, $\frac{H}{16} \times \frac{W}{16} \times 4C$, and $\frac{H}{32} \times \frac{W}{32} \times 8C$, where H, W and C denote the feature height, width and channel width, respectively.

Table 1
Architecture configurations of UnravelNet.

Stage	Downs. rate	Model layers	Number of input/output channels	
			Tiny	Small
1	$\frac{H}{4} \times \frac{W}{4}$	*Stem Layer* $[Unravel\ Block] \times N_1$	3/32 32/32	3/64 64/64
2	$\frac{H}{8} \times \frac{W}{8}$	*DRFD Module* $[Unravel\ Block] \times N_2$	32/64 64/64	64/128 128/128
3	$\frac{H}{16} \times \frac{W}{16}$	*DRFD Module* $[Unravel\ Block] \times N_3$	64/128 128/128	128/256 256/256
4	$\frac{H}{32} \times \frac{W}{32}$	*DRFD Module* $[Unravel\ Block] \times N_4$	128/256 256/256	256/512 512/512
SMRA Distance			[11, 11, 11, 11]	
Number of Block			$[N_1, N_2, N_3, N_4] = [1, 4, 4, 2]$	
Activation Function			*ReLU*	

downsampling modules, and Unravel blocks, which collaboratively process the input image across multiple levels.

The input image **I** is first passed through the stem layer, where a preliminary downsampling operation is performed using a convolutional layer. This layer is parameterized as $(\mathrm{I} = 3, \mathrm{O} = \mathrm{C}, \mathrm{K} = 4, \mathrm{S} = 4, \mathrm{P} = 0)$, where I, O, K, S, C and P denote the input channels, output channels, kernel size, stride, the number of channels and the padding size, respectively. The stem layer operation can be expressed as:

$$\mathrm{Stem} = \mathrm{Norm}(\mathrm{ConvD}_{2\mathrm{D}}^{4\times4}(\mathbf{I})), \tag{1}$$

where $\mathrm{ConvD}_{2\mathrm{D}}^{i\times i}(\cdot)$ represents a two-dimensional convolutional downsampling operation with a $i \times i$ kernel size.

After the stem layer, features are passed through a series of Unravel blocks, denoted as $\times N_i$, where N_i represents the number of Unravel blocks in the ith stage.

Each subsequent stage integrates downsampling modules alongside the Unravel blocks. The downsampling modules employed in stages 2, 3, and 4 utilize the deep robust feature downsampling (DRFD) module (Lu et al., 2023), which combines convolutional operations and max-pooling to preserve fine-grained details and small-scale features.

The output features generated by the four stages of the Unravel blocks are collectively fed into a decoder, such as Oriented R-CNN (Xie et al., 2021), for RSOD tasks. Detailed architectural configurations of UnravelNet backbone are summarized in Table 1.

3.2. Unravel block

The Unravel block, as illustrated in the top section of Fig. 4, processes the input feature map **X** through the Unravel module to produce an intermediate feature map **Y**. This is followed by a convolutional layer with parameters $(\mathrm{I} = \mathrm{C}, \mathrm{O} = 2\mathrm{C}, \mathrm{K} = 1, \mathrm{S} = 1, \mathrm{P} = 0)$, a normalization layer and an activation function (Act), resulting in feature map $\mathbf{Y}'$. Subsequently, $\mathbf{Y}'$ undergoes another convolution operation with parameters $(\mathrm{I} = 2\mathrm{C}, \mathrm{O} = \mathrm{C}, \mathrm{K} = 1, \mathrm{S} = 1, \mathrm{P} = 0)$, producing the feature map $\mathbf{Y}''$. Finally, the output feature map **out** is computed as:

$$\begin{aligned} \mathbf{Y} &= \mathrm{Unravel_module}(\mathbf{X}), \\ \mathbf{Y}'' &= \mathrm{Conv}_{2\mathrm{D}}^{1\times1}(\mathrm{Act}(\mathrm{Norm}(\mathrm{Conv}_{2\mathrm{D}}^{1\times1}(\mathbf{Y})))), \\ \mathbf{out} &= \mathbf{X} + \mathrm{Norm}(\mathrm{drop}(\mathbf{Y}'')), \end{aligned} \tag{2}$$

where $\mathrm{drop}(\cdot)$ represents a dropout operation, set to 0.1 for UnravelNet in this work.

The Unravel module serves as the cornerstone of feature extraction in UnravelNet, as shown in the bottom section of Fig. 4. It leverages the inherent redundancy of the input feature maps by partitioning them into four segments along the channel dimension without increasing the overall number of input channels. Each segment undergoes specialized feature extraction before being fused, which optimizes feature redundancy and strikes a balance between comprehensive feature extraction and computational efficiency. This design is effective in addressing the challenges of multi-scale and low-quality object recognition.

The Unravel module incorporates four attention mechanisms: point attention (PA), local attention (LA), sparse mid-range attention (SMRA) and low-quality feature enhancement attention (LFEA). PA, LA and SMRA operate on feature maps to capture multi-scale object features effectively, while LFEA enhances low-quality features.

Specifically, the input feature map $\mathbf{X} \in \mathbb{R}^{H\times W\times C}$ is divided into four equal segments $\{\mathbf{X_1}, \mathbf{X_2}, \mathbf{X_3}, \mathbf{X_4}\}$, where each segment $\mathbf{X_i} \in \mathbb{R}^{H\times W\times \frac{C}{4}}$. Here, **X** denotes the tensor, $\mathbb{R}^{H\times W\times C}$ represents the real number field, and H, W and C refer to the height, width and number of feature channels, respectively. The segments $\mathbf{X_1}$, $\mathbf{X_2}$, $\mathbf{X_3}$ and $\mathbf{X_4}$ are then passed through the PA, LA, SMRA and LFEA modules, respectively. The resulting output feature maps from these submodules are concatenated to form the final output.

3.2.1. Recognizing multi-scale objects

RS images typically include a wide variety of objects distributed across multiple scales, characterized by subtle differences and intricate details. This inherent complexity necessitates the enhancement of representational capabilities for multi-scale object features in computer vision tasks. Furthermore, prior knowledge of RS imagery indicates that

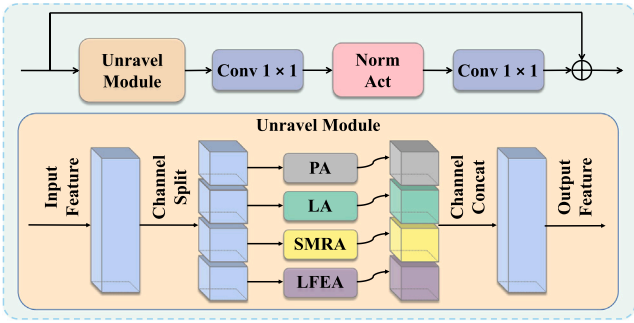

Fig. 4. Illustration of the Unravel block. The Unravel module splits the input features into four parts along the channel dimension, with each part processed by distinct attention modules. PA, LA, SMRA and LFEA stand for pointwise, local, sparse mid-range and low-quality feature enhancement attention, respectively.

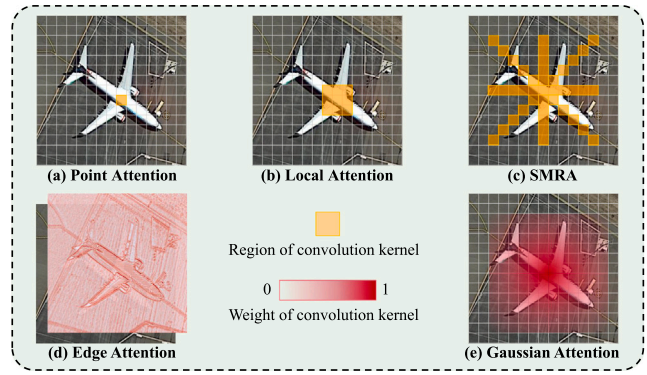

Fig. 5. Illustration of various attention modules. Yellow and red indicate the regions and weights of the convolutional kernel, respectively. Darker red corresponds to larger weights. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

these images generally lack large-scale objects, and the extraction of global information often incurs significant computational costs (this specific aspect is beyond the scope of the current study). Therefore, balancing performance improvement and computational efficiency is critical.

To address the challenges of multi-scale object feature extraction in RS images, this study introduces three multi-scale attention modules: PA, LA and SMRA. These modules are specifically designed to refine feature extraction across different object scales, thereby enhancing the overall accuracy of image analysis in computer vision applications.

PA: To effectively extract exceedingly small and detailed features, the Unravel module incorporates the PA submodule, as depicted in Fig. 5(a). The PA module leverages point attention to focus on extracting features from diminutive objects. Unlike traditional methods that may overlook or oversmooth such features, the PA module preserves subtle distinctions critical for small-scale object detection.

Initially, the input feature channels are expanded through a convolution with parameters $(\mathrm{I} = \mathrm{C}/4, \mathrm{O} = \mathrm{C}, \mathrm{K} = 1, \mathrm{S} = 1, \mathrm{P} = 0)$. This is followed by a normalization layer, activation function, and channel restoration via a convolution with $(\mathrm{I} = \mathrm{C}, \mathrm{O} = \mathrm{C}/4, \mathrm{K} = 1, \mathrm{S} = 1, \mathrm{P} = 0)$. A sigmoid operation is then applied to generate the attention map $\mathbf{A}_s$. The output feature $\mathbf{R_1}$ is computed as the sum of $\mathbf{X_1}$ and the element-wise product of $\mathbf{A}_s$ and $\mathbf{X_1}$:

$$\begin{aligned} &\mathbf{A_s} = \sigma(\mathrm{Conv}_{2\mathrm{D}}^{1\times1}(\mathrm{Norm}(\mathrm{Act}(\mathrm{Conv}_{2\mathrm{D}}^{1\times1}(\mathbf{X_1}))))), \\ &\mathrm{PA}(\mathrm{x}) = \mathbf{X_1} + \mathbf{X_1} \cdot \mathbf{A_s}, \end{aligned} \tag{3}$$

where $\sigma(\cdot)$ denotes the sigmoid function.

LA: To efficiently extract local features while maintaining lightweight constraints, the Unravel module integrates the LA submodule, as illustrated in Fig. 5(b). The LA module uses standard convolutional operations to excel at capturing precise local features.

The LA module processes $\mathbf{X_2}$ by applying a convolution with parameters $(\mathrm{I} = \mathrm{C}/4, \mathrm{O} = \mathrm{C}/4, \mathrm{K} = 3, \mathrm{S} = 1, \mathrm{P} = 1)$ to generate $\mathbf{X'_2}$. This is followed by a normalization layer and an activation function, producing the output feature $\mathbf{R_2}$.

SMRA: To manage medium-scale object features within lightweight frameworks, the Unravel module introduces the SMRA submodule, depicted in Fig. 5(c). This module incorporates contextual information to effectively preserve critical features of medium-sized objects, particularly those with irregular structural patterns.

The SMRA module processes the input feature x by employing a combination of horizontal, vertical, and oblique straight-line attention mechanisms to capture medium-range dependencies. A max-pooling operation with parameters $(K = 2, S = 2)$ reduces the feature size to $\mathbf{X'_3}$, enhancing the receptive field, resulting in $\mathbf{X_{3'}} \in \mathbb{R}^{\frac{H}{2}\times\frac{W}{2}\times\frac{C}{4}}$.

The processed feature generates the attention map $\mathbf{A}'_m$, integrating contextual information as follows:

$$\begin{aligned} \mathbf{A}_{ij} = &\sum_{n=0}^{n=\frac{L-1}{2}} \alpha_{(i\pm n)j} \cdot x_{(i\pm n)j} + \sum_{n=0}^{n=\frac{L-1}{2}} \alpha_{(i\pm n)(j\pm n)} \cdot x_{(i\pm n)(j\pm n)} \\ &+ \sum_{n=0}^{n=\frac{L-1}{2}} \alpha_{i(j\pm n)} \cdot x_{i(j\pm n)} + \sum_{n=0}^{n=\frac{L-1}{2}} \alpha_{(i\mp n)(j\pm n)} \cdot x_{(i\mp n)(j\pm n)}, \end{aligned} \tag{4}$$

where x_{ij} denotes the feature at position (i, j), and α_{ij} represents the learnable parameter associated with x_{ij}. In this study, L is set to 11. The enhanced attention map $\mathbf{A}'_m$ is resized via nearest-neighbor interpolation to match the size of $\mathbf{X_3}$, resulting in $\mathbf{A}_m \in \mathbb{R}^{H\times W\times\frac{C}{4}}$. Finally, the output feature $\mathbf{R_3}$ is computed as the element-wise product of $\mathbf{X_3}$ and $\mathbf{A}_m$: $\mathbf{R_3} = \mathbf{A}_m \cdot \mathbf{X_3}$.

3.2.2. Enhancing low-quality features

RS images often suffer from lower quality compared to natural images due to limitations in optical satellite resolution and the imaging distance. As a result, objects within these images can appear blurred or partially occluded, complicating feature extraction and reducing detection accuracy. These challenges adversely affect overall model performance.

To mitigate these issues, we propose the LFEA module, shown in Fig. 6. This module incorporates edge attention mechanisms and Gaussian distribution-based attention to enhance the feature representation of objects in RS imagery.

Stage 1: Edge Feature Enhancement

At the first stage, where the feature size is reduced to 1/4 of the original input, detailed features and edge information are preserved. To effectively extract edge features, we employ Scharr filters, which offer improved rotation invariance and reduced edge response errors compared to the Sobel operator. The Scharr filters are defined as follows:

$$\mathrm{S}_x = \begin{bmatrix} -3 & 0 & 3 \\ -10 & 0 & 10 \\ -3 & 0 & 3 \end{bmatrix}, \quad \mathrm{S}_y = \begin{bmatrix} -3 & -10 & -3 \\ 0 & 0 & 0 \\ 3 & 10 & 3 \end{bmatrix}. \tag{5}$$

The Scharr filters are applied via $\mathrm{Conv}_{2\mathrm{D}}^{s_x}(\cdot)$ and $\mathrm{Conv}_{2\mathrm{D}}^{s_y}(\cdot)$ convolution layers. Horizontal and vertical edge responses are computed, then combined using the Euclidean norm to generate the final edge map:

$$\mathrm{F}_{\mathrm{Edge}} = \sqrt{\left(\mathrm{Conv}_{2\mathrm{D}}^{s_x}(\mathbf{X_4})\right)^2 + \left(\mathrm{Conv}_{2\mathrm{D}}^{s_y}(\mathbf{X_4})\right)^2}, \tag{6}$$

which the s_x and s_y denote the parameters of convolution.

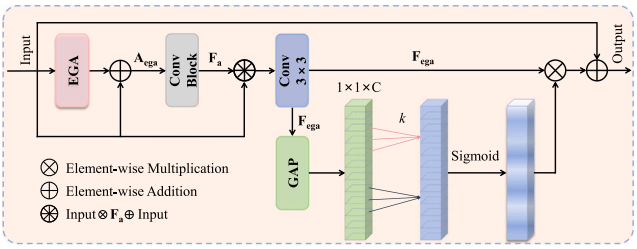

Fig. 6. Illustration of LFEA module. EGA refers to edge-Gaussian attention, employing edge attention or Gaussian attention at different stage to enhance feature representation. GAP and k denote global average pooling and convolutional kernel size, respectively.

The edge map is normalized, activated, and added back to the original feature map, creating an edge-enhanced feature representation. This enhanced map is further refined through a convolutional block.

Stages 2–4: Gaussian-Based Feature Enhancement

As the spatial size of feature maps decreases, edge information becomes less discernible, while object characteristics tend to exhibit a normal distribution. To better represent these objects, we employ Gaussian convolution kernels that maintain object characteristics and minimize information loss during network propagation.

A fixed Gaussian kernel is utilized to enhance salient and uncertainty features. This filter assigns higher weights to neighboring pixels closer to its center, emphasizing key features and suppressing background noise. Consequently, this enhances the network's focus on the most informative regions without introducing artifacts from high-frequency noise. The Gaussian filter is defined as:

$$G_{(i,j)} = \frac{1}{2\pi\sigma^2} e^{-\frac{i^2+j^2}{2\sigma^2}}, \tag{7}$$

where σ is the standard deviation, controlling the smoothing extent. i and j represent the spatial coordinates in the two-dimensional feature map. The Gaussian kernel is applied via depthwise convolution, ensuring channel-wise independence of features.

The Gaussian-based feature map is normalized and activated to suppress noise while retaining informative regions:

$$F_{Gaussian} = \mathrm{Act}\left(\mathrm{Norm}\left(\mathrm{Conv}_{2D}^{G_{(i,j)}}(\mathbf{X_4})\right)\right). \tag{8}$$

Integration and Final Feature Representation

As shown in Fig. 6. The LFEA module processes the input feature $\mathbf{X_4}$ by integrating either edge attention in Stage 1 or Gaussian attention in Stages 2–4, as illustrated in Fig. 5(d, e). These output features are added with the input features to form an attention map $\mathbf{A_{ega}}$. A convolutional block (Conv_Block) further extracts refined features, producing the intermediate output $\mathbf{F_a}$:

$$\begin{aligned} \mathbf{A_{ega}} &= \mathbf{X_4} \oplus \mathrm{EGA}(\mathbf{X_4}), \\ \mathbf{F_a} &= \mathrm{Conv_Block}(\mathbf{A_{ega}}), \end{aligned} \tag{9}$$

where EGA refers to edge-Gaussian attention, which employs stage-specific attention mechanisms: edge attention for Stage 1 and Gaussian attention for Stages 2–4. $\oplus$ denotes element-wise addition. Conv_Block includes an initial convolution layer mapping input channels to 64 intermediate channels, followed by batch normalization and an activation layer (e.g., ReLU) for non-linearity. Two further convolution layers with parameters ($K = 3, S = 1$) expand the features to 256 channels. This setup is succeeded by a sequence of normalization and activation layers, ending with a final convolution layer that returns the output to the original channels, preserving spatial dimensions.

An element-wise multiplication between $\mathbf{X_4}$ and $\mathbf{F_a}$, followed by a skip connection and a 3×3 convolution, yields the initial fused features $\mathbf{F_{ega}}$:

$$\mathbf{F_{ega}} = \mathrm{Conv}_{2D}^{3\times3}\left((\mathbf{X_4} \otimes \mathbf{F_a}) \oplus \mathbf{X_4}\right), \tag{10}$$

where $\otimes$ denotes element-wise multiplication.

Inspired by Wang et al. (2020), local attention is introduced to highlight critical feature channels. Global average pooling (GAP) aggregates $\mathbf{F_{ega}}$, and channel attention weights are computed using a one-dimensional convolution followed by a sigmoid activation:

$$\mathbf{R_4} = \mathrm{Norm}\left(\mathbf{X_4} \oplus \sigma\left(\mathrm{Conv}_{1D}^{k}\left(\mathrm{GAP}(\mathbf{F_{ega}})\right) \otimes \mathbf{F_{ega}}\right)\right), \tag{11}$$

where $\sigma(\cdot)$ denotes the sigmoid function. $\mathrm{Conv}_{1D}^{k}(\cdot)$ represents the one-dimensional convolution with kernel size k. The kernel size k is adaptively set as $k = \left|\frac{1+\log_2(C)}{2}\right|_{odd}$ based on the number of channels C, ensuring proportionality. In this study, the value of k was set identically to the approach adopted by Wang et al. (2020).

This attention strategy highlights critical channels while suppressing redundant channels and noise, thereby enhancing the semantic representation of the features.

In summary, the LFEA module combines edge detection and Gaussian attention to enhance salient and uncertainty features, significantly improving the robustness of feature representation in low-quality RS images.

4. Experiments

This section presents the datasets used, training configurations, and evaluation strategies that ensure reproducibility and a clear understanding of the testing scenarios. Then, we present the experimental results of UnravelNet on the outlined datasets. Next, we conduct several ablation studies to analyze the contribution of different components of our method. Finally, we provide visualizations to better illustrate the strengths of UnravelNet in comparison to existing methods.

4.1. Datasets

We evaluate the proposed method on four publicly available RSOD datasets: DOTA-v1.0 (Xia et al., 2018), DOTA-v1.5 (Xia et al., 2018), DIOR-R (Cheng et al., 2022a), and FAIR1M-v1.0 (Sun et al., 2022).

DOTA-v1.0 is an extensive dataset of aerial images consisting of 2,806 pictures with resolutions ranging from 800×800 to 4000×4000 pixels. It features 188,282 objects distributed across 15 different categories which include basketball court (BC), large vehicle (LV), baseball diamond (BD), harbor (HA), bridge (BR), ground track field (GTF), helicopter (HC), small vehicle (SV), ship (SH), plane (PL), tennis court (TC), storage tank (ST), soccer ball field (SBF), roundabout (RA), and swimming pool (SP). This dataset is segmented into three parts: training, validation, and testing, containing 1,411, 458, and 937 images respectively.

DOTA-v1.5 builds upon DOTA-v1.0 by introducing a new category, container crane (CC), and annotating tiny instances smaller than 10 pixels. It contains 403,318 instances, while retaining the same images, number of samples, and dataset splits as DOTA-v1.0.

DIOR-R is derived from the DIOR dataset (Li et al., 2020) and consists of 23,463 images with 192,518 instances spanning 20 common object classes. Unlike DIOR, each instance is annotated with an oriented bounding box. The training and validation subsets include 11,725 images, while the testing subset comprises 11,738 images.

FAIR1M-v1.0 is a recently released dataset targeting fine-grained object detection in RS images. It features over 40,000 images and more than 1 million instances distributed across 37 fine-grained categories. The training and testing subsets consist of 16,488 and 8,137 images, respectively.

Table 2
Experimental results on DOTA-v1.0 test set (Xia et al., 2018) under single-scale training and testing setting. UnravelNet-S is built within the framework of Oriented RCNN (Xie et al., 2021). The items with red and blue colors indicate the best and second-best results of each column, respectively.

Method	#P	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	**mAP**
DETR-based:																	
O^2-DETR (Ma et al., 2021)	–	86.01	75.92	46.02	66.65	79.70	79.93	**89.17**	90.44	81.19	76.00	56.91	62.45	64.22	65.80	58.96	72.15
AO^2-DETR (Dai et al., 2022)	40.8M	89.27	84.97	56.67	74.89	78.87	82.73	87.35	90.50	84.68	85.41	61.97	69.96	74.86	72.39	71.62	77.73
ARS-DETR (Zeng et al., 2024)	41.6M	86.97	75.56	48.32	69.20	77.92	77.94	87.69	90.50	77.31	82.86	60.28	64.58	74.88	71.76	66.62	74.16
PSD-SQ (Feng and Wang, 2024)	–	89.67	85.43	**57.28**	75.24	79.98	81.21	88.30	89.88	**88.08**	86.35	**69.63**	68.45	75.32	69.46	**72.65**	78.46
One-stage:																	
Rotated FCOS (Tian et al., 2020)	31.9M	88.52	77.54	47.06	63.78	**80.42**	80.50	87.34	90.39	77.83	84.13	55.45	65.84	66.02	72.77	49.17	72.45
R^3Det (Yang et al., 2021a)	41.9M	89.00	75.60	46.64	67.09	76.18	73.40	79.02	90.88	78.62	84.88	59.00	61.16	63.65	62.39	37.94	69.70
S^2ANet (Han et al., 2021a)	38.5M	89.11	82.84	48.37	71.11	78.11	78.39	87.25	90.83	84.90	85.64	60.36	62.60	65.26	69.13	57.94	74.12
GWD (Yang et al., 2021b)	41.9M	88.82	82.94	55.63	72.75	78.52	83.10	87.46	90.21	86.36	85.44	64.70	61.41	73.46	76.94	57.38	76.34
KLD (Yang et al., 2021c)	41.9M	88.90	84.17	55.80	69.35	78.72	84.08	87.00	89.75	84.32	85.73	64.74	61.80	76.62	**78.49**	70.89	77.36
SASM (Hou et al., 2022)	36.6M	86.42	78.97	52.47	69.84	77.30	75.99	86.72	**90.89**	82.63	85.66	60.13	68.25	73.98	72.22	62.37	74.92
O-RepPoints (Li et al., 2022)	36.6M	87.02	83.17	54.13	71.16	80.18	78.40	87.28	**90.90**	85.97	86.25	59.90	**70.49**	73.53	72.27	58.97	75.97
TIR-Net (Li et al., 2024)	40.0M	89.26	82.42	53.57	72.23	79.15	78.87	86.96	90.87	84.25	84.76	65.43	68.50	67.60	70.35	54.22	75.23
SOOD (Xi et al., 2024)	–	89.20	84.54	51.50	72.75	74.78	77.70	87.77	90.88	85.12	85.04	65.76	65.63	73.27	71.03	57.95	75.53
Two-stage:																	
RoI Trans. (Ding et al., 2019)	55.1M	89.01	77.48	51.64	72.07	74.43	77.55	87.76	90.81	79.71	85.27	58.36	64.11	76.50	71.99	54.06	74.05
Gliding Vertex (Xu et al., 2020)	41.1M	89.64	85.00	52.26	77.34	73.01	73.14	86.82	90.74	79.02	**86.81**	59.55	**70.91**	72.94	70.86	57.32	75.02
CenterMap (Wang et al., 2021a)	41.1M	89.02	80.56	49.41	61.98	77.99	74.19	83.74	89.44	78.01	83.52	47.64	65.93	63.68	67.07	61.59	71.59
O-RCNN (Xie et al., 2021)	41.1M	89.46	82.12	54.78	70.86	78.93	83.00	88.20	**90.90**	87.50	84.68	63.97	67.69	74.94	68.84	52.28	75.87
ReDet (Han et al., 2021b)	31.6M	88.79	82.64	53.97	74.00	78.13	84.06	88.04	**90.89**	87.78	85.75	61.76	60.39	75.96	68.07	63.59	76.25
QPDet (Yao et al., 2022)	41.1M	89.55	83.66	54.06	73.93	78.93	83.08	88.29	**90.89**	86.60	84.80	62.03	65.55	74.16	70.09	58.16	76.25
RPGAOD (Qiao et al., 2023)	41.1M	89.34	83.53	54.39	76.59	78.09	81.44	87.64	90.83	85.89	85.33	65.44	64.96	73.73	70.31	59.50	76.47
ARC-R50 (Pu et al., 2023)	74.4M	89.40	82.48	55.33	73.88	79.37	84.05	88.06	**90.90**	86.44	84.83	63.63	70.32	74.29	71.91	65.43	77.35
LSKNet-S (Li et al., 2023b)	31.0M	89.66	**85.52**	**57.72**	75.70	74.95	78.69	88.24	90.88	86.79	86.38	66.92	63.77	**77.77**	74.47	64.82	77.49
PKINet-S (Cai et al., 2024)	**30.8M**	**89.72**	84.20	55.81	77.63	80.25	84.45	88.12	90.88	87.57	86.07	66.86	70.23	77.47	73.62	62.94	78.39
UnravelNet-T (Ours)	***26.7M***	*89.55*	*85.30*	*56.28*	***78.37***	*79.32*	***85.43***	*88.17*	*90.88*	***88.41***	*86.35*	*65.57*	*67.76*	*77.73*	*73.42*	*65.09*	***78.51***
UnravelNet-S (Ours)	*31.1M*	***89.71***	***86.08***	*55.50*	***77.93***	***80.85***	***86.21***	***88.36***	*90.88*	*86.50*	***86.62***	***70.44***	*68.00*	***78.36***	***80.56***	***73.82***	***79.99***

4.2. Experimental settings

The UnravelNet backbone was pre-trained on ImageNet-1K (Deng et al., 2009) for 300 epochs, following standard practices (Pu et al., 2023; Li et al., 2023b; Cai et al., 2024). For RSOD tasks, the model was implemented using the MMRotate (Zhou et al., 2022) toolbox and integrated within the Oriented RCNN framework (Xie et al., 2021). The training process spanned 36 epochs, utilizing the AdamW optimizer (Kingma, 2014) with an initial learning rate of 0.00015, momentum of 0.9, and weight decay of 0.05. A linear warm-up strategy was applied during the first 500 iterations, and the learning rate was reduced by a factor of 10 at epochs 24 and 33. Random horizontal and vertical flipping were employed for data augmentation to mitigate overfitting. All experiments were conducted on two NVIDIA RTX 3090 GPUs with a total batch size of 8.

For DOTA-v1.0, DOTA-v1.5, and FAIR1M-v1.0, consistent with prior works (Xie et al., 2021; Li et al., 2023b), the images were cropped into 1024×1024 patches with a stride of 824. The training and validation subsets were used for training, and the test set was evaluated using the online server. For the DIOR-R dataset, the images retained their original resolution of 800×800 pixels, and the same learning rate schedule as DOTA-v1.0 was applied.

4.3. Experimental results

4.3.1. Results on DOTA-v1.0

We evaluate UnravelNet against several SOTA approaches on the DOTA-v1.0 test set (Xia et al., 2018) using single-scale training and testing, with results summarized in Table 2. Remarkably, incorporating UnravelNet as the backbone leads to substantial mAP improvements, consistently outperforming ResNet-50 (He et al., 2016) and other RSOD-specific backbones, including ARC (Pu et al., 2023), LSKNet (Li et al., 2023b), and PKINet (Cai et al., 2024). When integrated with Oriented RCNN (Xie et al., 2021) decoder, UnravelNet achieves SOTA performance, with UnravelNet-T and UnravelNet-S attaining mAPs of 78.51% and 79.99%, respectively. These results surpass the previously best-performing method, PKINet (Cai et al., 2024), with significant gains in low-quality categories such as SP (+6.94%) and HC (+10.88%). Moreover, among the 15 common categories in the dataset, our model achieves the best or second-best in 11 of them, which further demonstrates that UnravelNet has strong generalization ability and can significantly improve detection accuracy.

Table 3
Comparison of different backbones with Oriented RCNN (Xie et al., 2021) framework on DOTA-v1.0 test set (Xia et al., 2018). Parameters are computed for backbones only.

Backbone	#P	**mAP**
ResNet-18 (He et al., 2016)	**11.2M**	74.20
ResNet-50 (He et al., 2016)	23.3M	75.87
IMP-Swin-T (Wang et al., 2022)	44.8M	76.07
ARC-R50 (Pu et al., 2023)	74.4M	77.35
LSKNet-S (Li et al., 2023b)	31.0M	77.49
PKINet-S (Cai et al., 2024)	13.7M	78.39
*UnravelNet-T (**ours**)*	***9.6M***	***78.51***
*UnravelNet-S (**ours**)*	*14.0M*	***79.99***

As shown in Table 3, we also compared UnravelNet with other backbones on the DOTA-v1.0 test set using the Oriented RCNN framework. UnravelNet-T reduces parameters by 14.3% compared to ResNet-18 while achieving a 4.31% higher mAP. Similarly, UnravelNet-S uses only 60.09% of the parameters of ResNet-50 but outperforms it by 4.12%. Additionally, UnravelNet-S surpasses PKINet-S by 1.50% with a comparable parameters, achieving the best performance among all backbones evaluated.

To further evaluate the versatility of UnravelNet, we tested its integration with various RSOD frameworks. Here, the training and testing strategies we adopted are consistent with those in Section 4.2.

Table 4 demonstrates the superiority of UnravelNet-S over the ResNet-50 baseline across seven SOTA detection frameworks on the DOTA-v1.0 benchmark. With a significant 40% parameter reduction (14.0M vs. 23.3M), our method consistently surpasses the baseline by margins ranging from 1.70% to 8.70% mAP points. Notably, when integrated with R^3Det (Yang et al., 2021a), UnravelNet-S achieves the most substantial improvement of +8.70% mAP (78.40% vs. 69.70%), demonstrating its strong compatibility with advanced rotational detection paradigms. The results particularly highlight UnravelNet's effectiveness in handling complex orientation variations prevalent in RS imagery, as evidenced by consistent gains across all detection frameworks (mean improvement: +3.71% mAP).

Table 4
Comparison of mAP with different detection frameworks on the DOTA-v1.0 test set (Xia et al., 2018). Parameters are computed for backbones only.

Framework	ResNet-50 (He et al., 2016)	*UnravelNet-S*
#P	23.3M	*14.0M (−**40%**)*
FR-O (Ren et al., 2016)	73.17	*76.61 (+**3.44**)*
Roi Trans. (Ding et al., 2019)	74.05	*77.38 (+**3.33**)*
R-FCOS (Tian et al., 2020)	72.45	*74.59 (+**2.14**)*
R^3Det (Yang et al., 2021a)	69.70	*78.40 (+**8.70**)*
S^2ANet (Han et al., 2021a)	74.12	*78.52 (+**4.40**)*
QPDet (Yao et al., 2022)	76.25	*77.95 (+**1.70**)*
KFIoU (Yang et al., 2023)	75.10	*77.83 (+**2.73**)*

4.3.2. Results on DOTA-v1.5

The experimental results on the more challenging DOTA-v1.5 test set (Xia et al., 2018), as shown in Table 5, demonstrate the superior performance of UnravelNet compared to SOTA methods under single-scale training and testing conditions. UnravelNet-S achieves an impressive mAP of 73.09%, surpassing all other listed methods. Compared to baseline models (e.g., FR-O (Ren et al., 2016) and RetinaNet-O (Lin et al., 2017)) and more advanced methods (e.g., LSKNet-S (Li et al., 2023b), SOOD (Xi et al., 2024), and PKINet-S (Cai et al., 2024)), UnravelNet-S demonstrates superior performance across multiple categories, securing the highest AP scores in BR (55.67%), LV (81.70%), SH (88.99%), BC (85.93%), SBF (68.51%), RA (74.67%), and HC (72.44%). Additionally, it achieves competitive second-best performance in PL (80.08%), TC (90.85%), HA (77.35%), and CC (37.98%). These results highlight the robustness of UnravelNet-S in detecting objects with diverse scales, orientations, and contextual complexities. Compared to its closest competitor, PKINet-S (71.47%), UnravelNet-S achieves a significant 1.62% improvement in mAP, further demonstrating the effectiveness of UnravelNet in the benchmark evaluation.

4.3.3. Results on DIOR-R

The comparative evaluation on the DIOR-R test set (Cheng et al., 2022a) further validates the remarkable performance and robustness across datasets of UnravelNet, as shown in Table 6. Integrated into the Oriented RCNN (Xie et al., 2021) framework, UnravelNet-S achieves an mAP of 69.48%, significantly outperforming other competitive methods. Compared with the baseline Oriented RCNN based on ResNet-50 (He et al., 2016), UnravelNet-S shows an improvement of 5.66% (69.48% vs. 63.82%), which fully demonstrates the effectiveness of UnravelNet in enhancing detection capabilities. Among other notable methods using the ResNet-50 backbone, UnravelNet-S also performs outstandingly: it outperforms RoI Trans. (Ding et al., 2019) by 5.61% (69.48% vs. 63.87%), AOPG (Cheng et al., 2022a) by 5.07% (69.48% vs. 64.41%), DODet (Cheng et al., 2022b) by 4.38% (69.48% vs. 65.10%), DCFL (Xu et al., 2023) (69.48% vs. 66.80%) and Oriented Rep (Xie et al., 2021) (69.48% vs. 66.71%). Within the Oriented RCNN framework, UnravelNet-S also demonstrates its robustness by outperforming other alternative backbones. It exceeds LSKNet-S (Li et al., 2023b) (69.48% vs. 65.90%), PKINet-S (Cai et al., 2024) (69.48% vs. 67.03%), and DecoupleNet-D2 (Lu et al., 2024) (69.48% vs. 67.08%). The consistent performance advantage across various methods indicates that UnravelNet effectively addresses the challenges posed by the DIOR-R dataset.

4.3.4. Results on FAIR1M-v1.0

The FAIR1M-v1.0 dataset (Sun et al., 2022), a benchmark for fine-grained RSOD with high inter-class similarity, serves as another evaluation platform. As shown in Table 7, UnravelNet demonstrates significant improvements over a variety of established methods. Specifically, it outperforms RoI Trans. (Ding et al., 2019) by 5.95%, Gliding Vertex (Xu et al., 2020) by 8.58%, and also exceeds the baseline Oriented RCNN (Xie et al., 2021) by 5.59%. Compared with recent SOTA methods, UnravelNet-S still maintains its lead, surpassing TIOE-Det (Ming et al., 2023) by 9.28%, PETDet (Li et al., 2023a) by 1.48%, and PCLDet (Ouyang et al., 2023) by 0.94%. These improvements highlight the strong capability of UnravelNet in handling fine-grained RSOD tasks.

4.3.5. Results on COCO

To assess the generality of UnravelNet across both RS and natural images, we evaluate its performance on the COCO dataset (Lin et al., 2014), as detailed in Table 8. UnravelNet-S achieves an mAP of 44.0% within the Mask R-CNN (He et al., 2017) framework, outperforming all compared backbones. Although it was initially designed for RSOD, it performs excellently in natural image scenes as well. UnravelNet-S exhibits an exceptional balance between parameters efficiency and detection performance. With a parameters of 14.0M, it surpasses ResNet-18 (He et al., 2016) (11.2M, 34.0% mAP) by a significant margin of 10.0% in mAP, despite a modest increase in parameters. Furthermore, in contrast to FasterNet-S (Chen et al., 2023a) (28.9M, 39.9% mAP), UnravelNet-S achieves a 4.1% mAP improvement with approximately half the parameters, demonstrating superior resource utilization. Even compared to the previous best-performing method, UnravelNet-S outperforms PKINet-S (Cai et al., 2024) (13.7M, 43.4% mAP) by 0.6% in mAP with a negligible parameter increase of 0.3M, reinforcing its competitive edge in lightweight design.

Due to the relatively limited presence of multi-scale variations and low-quality features in the COCO dataset compared to the RSOD datasets, the intrinsic strengths of our proposed model are not fully leveraged in this experimental context. The COCO benchmark evaluations primarily aim to validate the cross-domain generalizability of UnravelNet. While the quantitative metrics obtained on this dataset may not achieve SOTA performance, our framework demonstrates superior detection accuracy over multiple established backbone architectures in comparative analyses.

4.4. Visualization

Fig. 7 illustrates the visualization results of different models on the DOTA-v1.0 test set (Xia et al., 2018), a widely used large-scale benchmark for RSOD. The visualizations for ResNet-50 (He et al., 2016), PKINet-S (Cai et al., 2024), and our UnravelNet-S were generated using the MMRotate (Zhou et al., 2022) toolbox, while ARC-R50 (Li et al., 2023b) visualizations were based on the MMDetection (Chen et al., 2019b) toolbox.

ResNet-50 fails to extract sufficient contextual information, leading to misclassification, such as detecting harbors as planes due to visual similarity. Models like ARC and PKINet, which lack point attention modules, also fail to fully capture the details of objects, resulting in missed detections. In contrast, our method employs a multi-branch attention scheme, effectively capturing contextual information and addressing small (e.g., SV), medium (e.g., BR, SH), and large objects (e.g., GTF, SBF). By incorporating edge information and Gaussian distribution, our model demonstrates enhanced sensitivity to edge details, achieving more precise localization than competing methods.

Table 5
Experimental results on the DOTA-v1.5 test set (Xia et al., 2018) compared with SOTA methods using single-scale training and testing. The UnravelNet-S backbone is implemented within the Oriented RCNN framework (Xie et al., 2021).

Method	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	CC	mAP
FR-O (Ren et al., 2016)	71.89	74.47	44.45	59.87	51.28	68.98	79.37	90.78	77.38	67.50	47.75	69.72	61.22	65.28	60.47	1.54	62.00
RetinaNet-O (Lin et al., 2017)	71.43	77.64	42.12	64.65	44.53	56.79	73.31	90.84	76.02	59.96	46.95	69.24	59.65	64.52	48.06	0.83	59.16
Mask R-CNN (He et al., 2017)	76.84	73.51	49.90	57.80	51.31	71.34	79.75	90.46	74.21	66.07	46.21	70.61	63.07	64.46	57.81	9.42	62.67
HTC (Chen et al., 2019a)	77.80	73.67	51.40	63.99	51.54	73.31	80.31	90.48	75.12	67.34	48.51	70.63	64.84	64.48	55.87	5.15	63.40
ReDet (Han et al., 2021b)	79.20	82.81	51.92	71.41	52.38	75.73	80.92	90.83	75.81	68.64	49.29	72.03	73.36	70.55	63.33	11.53	66.86
AO^2-DETR (Dai et al., 2022)	79.55	78.14	42.41	61.23	**55.34**	74.50	79.57	90.64	74.76	**77.58**	53.56	66.91	58.56	73.11	**69.64**	24.71	66.26
FSDet (Yu et al., 2022)	**80.81**	81.22	51.35	72.89	54.53	74.12	81.07	90.81	78.40	**76.85**	52.81	70.42	73.73	71.89	68.45	14.67	68.37
LSKNet-S (Li et al., 2023b)	72.05	**84.94**	55.41	**74.93**	52.42	**77.45**	81.17	**90.85**	79.44	69.00	62.10	**73.72**	**77.49**	**75.29**	55.81	**42.19**	70.26
SOOD (Xi et al., 2024)	**80.32**	84.41	52.59	**74.77**	**58.48**	76.90	86.97	**90.87**	78.62	76.56	62.93	71.16	74.64	**76.04**	55.97	25.09	70.39
PKINet-S (Cai et al., 2024)	80.31	**85.00**	**55.61**	74.38	52.41	76.85	**88.38**	**90.87**	**79.04**	68.78	**67.47**	72.45	76.24	74.53	64.07	37.13	**71.47**
*UnravelNet-S (**ours**)*	*80.08*	*84.82*	***55.67***	*73.95*	*52.56*	***81.70***	***88.99***	***90.85***	***85.93***	*69.39*	***68.51***	***74.67***	***77.35***	*74.61*	***72.44***	***37.98***	***73.09***

Table 6
Experimental results on DIOR-R test set (Li et al., 2020) compared with SOTA methods.

Method	Backbone	**mAP**
RoI Trans. (Ding et al., 2019)	ResNet-50 (He et al., 2016)	63.87
Oriented Rep (Xie et al., 2021)	ResNet-50 (He et al., 2016)	66.71
AOPG (Cheng et al., 2022a)	ResNet-50 (He et al., 2016)	64.41
DODet (Cheng et al., 2022b)	ResNet-50 (He et al., 2016)	65.10
GGHL (Huang et al., 2022)	ResNet-50 (He et al., 2016)	66.48
DCFL (Xu et al., 2023)	ResNet-50 (He et al., 2016)	66.80
SOOD (Xi et al., 2024)	ResNet-50 (He et al., 2016)	64.41
ORSR (Zheng et al., 2024)	ResNet-50 (He et al., 2016)	65.18
ARS-DETR (Zeng et al., 2024)	ResNet-50 (He et al., 2016)	66.12
O-RCNN (Xie et al., 2021)	ResNet-50 (He et al., 2016)	63.82
	LSKNet-S (Li et al., 2023b)	65.90
	PKINet-S (Cai et al., 2024)	67.03
	DecoupleNet-D2 (Lu et al., 2024)	67.08
	*UnravelNet-S (**ours**)*	***69.48***

Table 7
Experimental results on the FAIR1M-v1.0 test set (Sun et al., 2022) compared with SOTA methods using single-scale training and testing. The UnravelNet-S backbone is implemented within the Oriented RCNN framework (Xie et al., 2021, 2024).

Method	**mAP**
RoI Trans. (Ding et al., 2019)	38.49
G. V. (Xu et al., 2020)	35.86
DAL (Ming et al., 2021b)	29.00
R^3Det (Yang et al., 2021a)	31.07
CFC-Net (Ming et al., 2021a)	34.31
O-RCNN (Xie et al., 2021)	38.85
S^2ANet (Han et al., 2021a)	34.71
TIOE-Det (Ming et al., 2023)	35.16
SFRNet (Cheng et al., 2023)	40.74
PETDet (Li et al., 2023a)	42.96
PCLDet (Ouyang et al., 2023)	43.50
*UnravelNet-S (**ours**)*	***44.44***

Table 8
Experimental results on the COCO 2017 dataset (Lin et al., 2014). All models are implemented using the Mask R-CNN framework (He et al., 2017). Parameters are calculated based on an image size of 1280 × 800, considering only the backbones.

Method	#P	**mAP**
ResNet-18 (He et al., 2016)	**11.2**M	34.0
PVT-T (Wang et al., 2021b)	12.9M	36.7
ConvNeXt-N (Liu et al., 2022)	15.6M	41.3
FasterNet-S (Chen et al., 2023a)	28.9M	39.9
PKINet-S (Cai et al., 2024)	13.7M	43.4
*UnravelNet-S (**ours**)*	*14.0M*	***44.0***

4.5. Ablation study

To evaluate the effectiveness of the proposed UnravelNet, we conduct a series of ablation experiments on the DOTA-v1.0 test set (Xia et al., 2018). In these experiments, UnravelNet-T is adopted as the backbone network. It is integrated into the Oriented R-CNN (Xie et al., 2021) detector and pre-trained on the ImageNet-1K dataset (Deng et al., 2009) for 100 epochs to ensure fair comparison and stable convergence.

Each component in UnravelNet is designed to enhance the perception and representation of specific object characteristics. For instance, the PA module emphasizes fine-grained spatial details, while LFEA is tailored to enhance low-quality and ambiguous features.

In this ablation study, we isolate and evaluate the individual contributions of the PA, LA, SMRA, and LFEA modules. To eliminate potential interference from inter-branch interactions, each module is independently applied across the full feature channel space without any channel partitioning. This design allows for a more accurate assessment of each module's feature modeling capabilities. The results are summarized in Table 9. Additionally, we compare these single-module settings (first four rows) with various multi-module configurations (last four rows) to explore the synergy among modules.

The PA mechanism enhances the model's sensitivity to small regions, thereby improving the detection of small-scale objects. For example, the PA module achieves a strong result on the SV category, with an AP of 79.39%, which is the highest among all single-module configurations.

The LA and SMRA modules are both effective in capturing local context, but they differ in their focus. LA improves detection of regular-shaped objects, such as TC, BC, and SBF. In contrast, SMRA shows

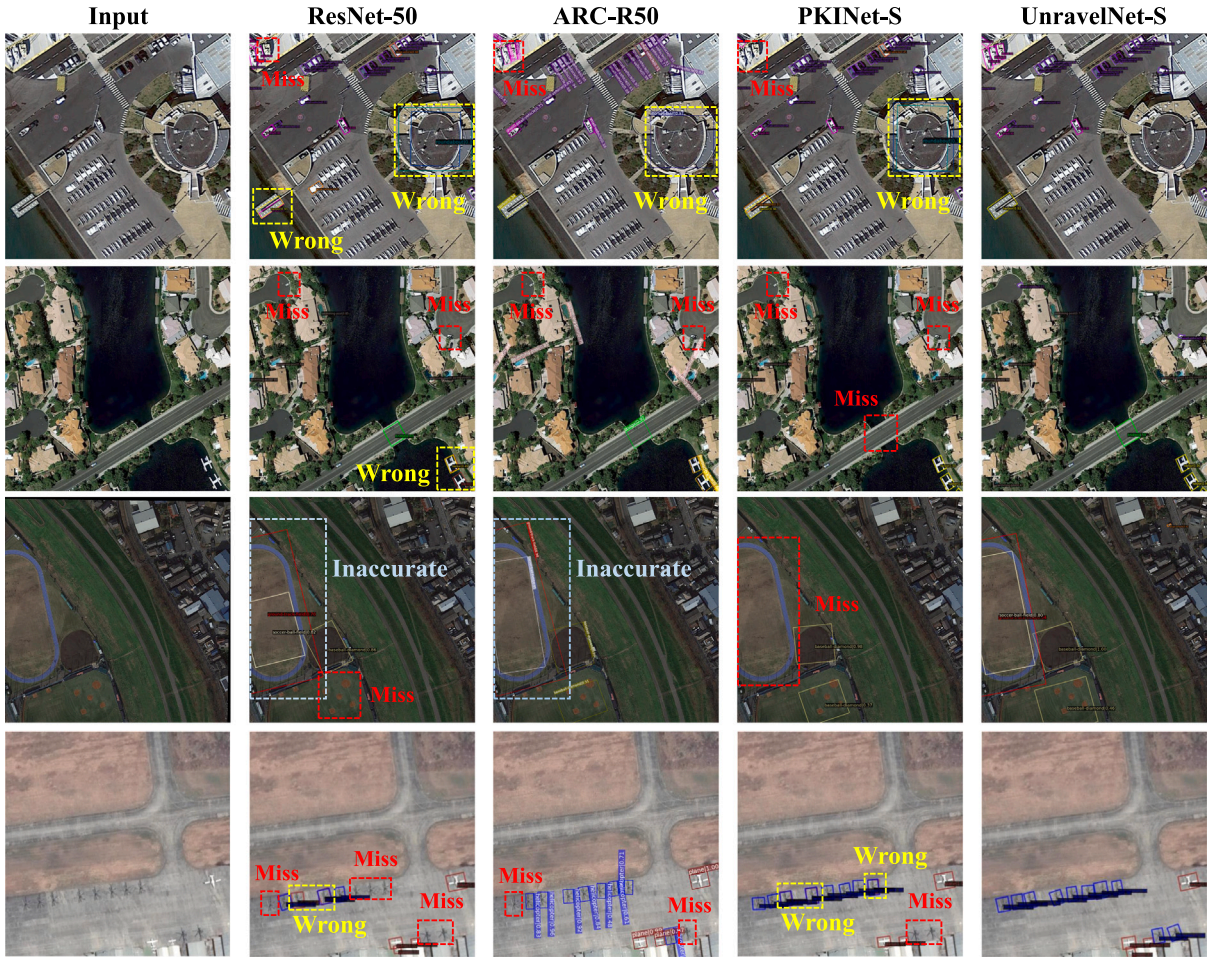

Fig. 7. Visualization of detection results on DOTA-v1.0 test set (Xia et al., 2018). The visualization results of ARC-R50 are based on the MMDetection (Chen et al., 2019b) toolbox, while ResNet-50, PKINet-S and UnravelNet-S visualization results are based on the MMRotate (Zhou et al., 2022) toolbox.

Table 9
Effectiveness of each component in the proposed UnravelNet. The mAP values are evaluated on the DOTA-v1.0 test set (Xia et al., 2018), and the parameters correspond to the backbone networks only. All backbones are pre-trained on the ImageNet-1K dataset (Deng et al., 2009) for 100 epochs and implemented within the Oriented RCNN framework (Xie et al., 2021). Each module utilizes comparable parameters (stem dims with 64, 64, 96, 16, 64, 78, 32 and 32, respectively).

PA	LA	SMRA	LFEA	#P	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	mAP
✓				11.0M	89.11	82.97	52.41	74.98	79.39	82.24	88.04	90.87	85.69	78.20	65.31	65.03	75.00	79.12	61.88	76.68
	✓			11.8M	89.43	84.28	52.38	77.82	78.15	84.25	88.11	**90.90**	87.09	86.04	**67.73**	62.22	76.74	77.90	55.78	77.25
		✓		9.8M	**89.80**	83.00	53.47	74.05	75.15	82.84	87.31	90.87	85.52	85.55	64.63	64.10	74.58	71.79	63.51	76.41
			✓	9.1M	89.45	84.70	54.45	76.90	77.71	84.31	88.35	90.88	86.79	**86.29**	62.78	64.03	76.94	78.64	61.99	77.61
✓	✓			7.8M	89.23	82.82	52.96	76.87	79.12	84.15	88.01	90.88	87.16	86.06	65.06	63.52	76.39	78.77	62.30	77.55
✓	✓	✓		8.7M	89.61	84.40	**55.90**	78.25	76.21	**85.17**	**88.43**	90.86	86.66	86.17	66.53	64.35	**77.84**	72.99	**64.86**	77.88
		✓	✓	9.9M	89.77	83.36	54.28	76.75	78.46	84.53	88.12	**90.90**	**87.19**	86.24	65.28	63.96	76.94	**79.54**	63.12	77.90
✓	✓	✓	✓	9.6M	89.59	**85.33**	55.72	**78.49**	**79.46**	**85.17**	87.85	90.88	86.74	86.09	66.76	**66.13**	77.09	73.66	63.43	**78.16**

superior performance on irregular and deformable objects such as PL and HC, benefiting from its multi-scale and relational modeling capabilities.

LFEA is specifically designed to enhance the representation of degraded features. By applying edge attention in shallow layers and Gaussian-based attention in deeper layers, it strengthens object boundaries and improves detection under challenging conditions such as blur or low contrast.

When combining PA and LA, UnravelNet-T achieves an mAP of 77.55%, leveraging the strengths of both modules. This configuration delivers competitive performance on small objects (e.g., SV: 79.12%) and regular structures (e.g., TC: 90.88%, BC: 87.16%).

Incorporating SMRA into the PA+LA setting further boosts the mAP to 77.88%. This three-module combination excels in handling complex shapes, with notable improvements in HC (64.86%), SH (88.43%), and BR (55.90%), indicating that SMRA effectively complements the detailed spatial modeling provided by PA and LA.

The dual-branch combination of SMRA and LFEA achieves a comparable mAP of 77.90%, demonstrating that SMRA's structural perception and LFEA's enhancement of ambiguous features can work effectively in tandem. This setup yields the best AP scores for TC (90.90%), BC (87.19%), and SP (79.54%) among all partial combinations.

Finally, the full integration of all four modules — PA, LA, SMRA, and LFEA — yields the highest overall mAP of 78.16%, confirming the complementary benefits of each component and the effectiveness of the proposed multi-branch design.

5. Conclusion

This paper introduces UnravelNet, a novel backbone network designed for robust RSOD across diverse and challenging datasets. By

integrating a multi-branch attention scheme combining PA, LA, SMRA and LFEA modules, UnravelNet enhances multi-scale feature extraction while maintaining parameters efficiency. This design effectively captures contextual information, improves localization precision, and strengthens feature representations, particularly in low-quality and multi-scale scenarios.

Comprehensive experiments validate the efficacy of UnravelNet across multiple benchmark datasets, including DOTA-v1.0, DOTA-v1.5, DIOR-R, FAIR1M-v1.0 and COCO. On DOTA-v1.0, UnravelNet demonstrates its superiority by achieving SOTA performance, significantly surpassing traditional backbones such as ResNet-50 and RSOD-specific designs like ARC, LSKNet and PKINet. Its ability to generalize is further supported by consistent improvements observed on other datasets.

The proposed UnravelNet achieves significant performance improvements with the smaller parameters, underscoring its potential for real-world deployment. Its robustness across both RS and natural image datasets highlights its adaptability and broad application potential.

Future work will explore extending UnravelNet to other CV tasks, such as semantic segmentation and change detection, as well as optimizing its deployment on edge devices for real-time RS applications. These efforts aim to further expand its utility while adhering to the principles of computational efficiency and high accuracy.

CRediT authorship contribution statement

Wei Lu: Writing – review & editing, Visualization, Validation, Methodology. **Hui-Dong Li:** Writing – original draft, Visualization, Validation. **Chao Wang:** Writing – review & editing. **Si-Bao Chen:** Writing – review & editing, Funding acquisition, Conceptualization. **Chris H.Q. Ding:** Resources, Funding acquisition. **Jin Tang:** Writing – review & editing, Supervision, Resources, Funding acquisition. **Bin Luo:** Writing – review & editing, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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